



## **BRIGMAN**

### **Deep Sea Technology**

1600 Pennsylvania Ave NW  
Washington DC  
20500, United State

# **Requirements Analysis Report (BCIC)**

Vendor: **Brigman Deep Sea Technology**

**Prepared by:**

Muhammad Ridwan Putra Jasni

Kannan s/o Rajamohan

Chng Zong Han

Lin Yuhao

Muhammad Mikhail Bin Mazlan

*Date:* SUBMISSION\_DATE



@brigman\_deepsea



www.brigman.com



info@brigman.com



123-456-7890



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20500, United State

## 1 Purpose of This Document

This document records the formal requirements analysis performed by Brigman Deep Sea Technology on the Project Specification (PS) for the Abyss Navigator Suite. Following the BCIC (Breakdown, Clarify, Interpret, Categorize) methodology, this analysis identifies and resolves incongruities, clarifies ambiguities, and prepares requirements for subsequent Software Requirements Specification (SRS) development. This report represents the output of the Analysing method within the Requirements Engineering framework.

## 2 Executive Summary

This report presents the findings from Brigman Deep Sea Technology's analysis of the Project Specification (PS) for the **Abyss Navigator Suite**. Our systematic BCIC approach has identified critical gaps and ambiguities that require clarification to ensure successful implementation of the unmanned, autonomous deep-sea submersible control system.

Key analysis outcomes include:

- **Communication Specification Gap:** Quantitative definitions for bandwidth, latency, and intermittent connectivity thresholds are absent.
- **Safety Protocol Deficiency:** Emergency handling lacks operational procedures, escalation protocols, and defined threshold values.
- **Interface Definition Shortfall:** Sensor integration requirements omit interface protocols, data formats, and power specifications.
- **Autonomy Level Ambiguity:** Unmanned operations context requires clearer definition of autonomy levels and remote intervention conditions.
- **Data Management Specificity:** Traceability requirements need concrete data retention periods and format standards.





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### Requirements Analysis (BCIC)

Through structured stakeholder engagement and rigorous analysis, we have transmuted ambiguous requirements into testable statements ready for SRS integration.

## 3 Vendor Analysis Team

The BCIC analysis was conducted by Brigman Deep Sea Technology's dedicated Requirements Analysis Team, comprising specialists from multiple engineering departments.

| Team Member                 | Employee ID | Department               | Primary Analysis Focus                         |
|-----------------------------|-------------|--------------------------|------------------------------------------------|
| Muhammad Ridwan Putra Jasni | BDST-RE-001 | Requirements Engineering | Analysis Methodology & Project Coordination    |
| Chng Zong Han               | BDST-SE-015 | Safety Engineering       | Emergency Protocols & Safety Systems           |
| Lin Yuhao                   | BDST-CS-022 | Communication Systems    | Bandwidth Constraints & Connectivity Protocols |
| Muhammad Mikhail Bin Mazlan | BDST-AI-029 | Autonomous Systems       | Autonomy Levels & Mission Execution            |
| Kannan s/o Rajamohan        | BDST-DM-036 | Data Management          | Logging Standards & Traceability Requirements  |

Table 1: Vendor Analysis Team Composition





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## 4 BCIC Analysis of Project Specification

As the selected vendor, Brigman Deep Sea Technology has applied the BCIC methodology to systematically analyze the Project Specification provided by Triton Deepwater. This section documents our application of the four BCIC activities.

### 4.1 Breakdown Activity

The Breakdown activity involved deconstructing the PS to identify incongruent, ambiguous, or missing requirements using our proprietary Incongruous Requirements Checklist (IRC), developed from ISO/IEEE standards and deep-sea operational expertise.





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| PS Reference | Incongruity Identified                                                                                       | Priority |
|--------------|--------------------------------------------------------------------------------------------------------------|----------|
| BO-L1-4      | No quantitative definition of "abnormal conditions" or threshold values for emergency response initiation    | Critical |
| IT-L0-C2     | "Constrained bandwidth" not quantified; no fallback mechanism specified for communication loss               | High     |
| Section 7.3  | Communication technology unspecified for deep-sea operational environments; missing performance requirements | Critical |
| BO-L1-6      | Traceability requirements lack specific data retention period and format standards                           | Medium   |
| Section 2.1  | Sensor integration mentioned but interface protocols, data formats, and power requirements unspecified       | High     |
| BO-L0-C1     | "Unmanned operations" context lacks clarification on autonomy levels and remote intervention protocols       | Medium   |

Table 2: Key Incongruities Identified During Breakdown

#### 4.1.1 Incongruous Requirements Checklist Application

Our IRC methodology examined requirements through four validation lenses:





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| Validation Category       | Applied Methodology                                                                              |
|---------------------------|--------------------------------------------------------------------------------------------------|
| Ambiguity Detection       | Flagged subjective terms ("appropriate", "secure", "constrained") for stakeholder clarification. |
| Testability Validation    | Assessed requirements for verifiability through testing, inspection, or analysis.                |
| Consistency Checking      | Cross-referenced requirements across PS sections for internal consistency.                       |
| Completeness Verification | Identified missing requirements based on operational context and industry standards.             |

Table 3: IRC Validation Categories Applied

## 4.2 Clarify Activity

We conducted structured stakeholder sessions with Triton Deepwater representatives to resolve identified incongruities. These sessions followed BCIC best practices, utilizing simplified visual models to facilitate mutual understanding.





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### Requirements Analysis (BCIC)

| Date       | Stakeholders                                         | Clarification Outcomes                                                                                                                                                                                                                                                                                          |
|------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-02-01 | Triton Safety Manager, Fleet Operations Lead         | <ul style="list-style-type: none"><li>Established quantitative definitions for abnormal conditions aligned with Section 6.4</li><li>Confirmed emergency protocols must maintain "Safety-First" policy consistency</li><li>Defined safety state transitions supporting mission continuity and recovery</li></ul> |
| 2026-02-03 | Triton IT Manager, Offshore Engineering Lead         | <ul style="list-style-type: none"><li>Quantified bandwidth constraints not exceeding 1 kbps with latency <math>\leq 30</math> seconds</li><li>Established interface standards for customer-supplied sensor integration</li><li>Defined fallback mechanisms for intermittent connectivity scenarios</li></ul>    |
| 2026-02-05 | Mission Logistics Manager, Safety Compliance Officer | <ul style="list-style-type: none"><li>Specified 7-year data retention period for operational accountability</li><li>Defined audit trail requirements supporting post-mission review</li><li>Confirmed AES-256 encryption for security and governance compliance</li></ul>                                       |

Table 4: Stakeholder Clarification Sessions

**Visual Model Utilization:** During clarification sessions, we employed:

- State transition diagrams illustrating emergency response workflows
- Bandwidth-latency operational envelope visualizations
- Interface data flow diagrams for sensor integration scenarios





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### 4.3 Interpret Activity

Based on clarification outcomes, we reinterpreted ambiguous PS requirements into precise, testable statements compliant with IEEE 830 standards.

#### Original Requirement: BO-L1-4 (Emergency Handling)

*"The business requires appropriate handling of abnormal or emergency conditions during missions."*

#### Interpreted Requirement: SYS-SAF-001

*"The system shall detect abnormal conditions exceeding predefined operational parameters within 5 seconds of onset. Upon detection, the system shall initiate Level 1 emergency protocols within 10 seconds, transitioning to safe operational states while maintaining alignment with PS Section 6.4 safety requirements."*

#### Original Requirement: IT-L0-C2 (Communication Constraint)

*"IT solutions shall operate under conditions of constrained bandwidth, latency, and intermittent connectivity."*

#### Interpreted Requirement: COM-REQ-001

*"The communication subsystem shall maintain mission-critical functionality under bandwidth constraints of 1kbps with maximum latency of 30 seconds. The system shall implement store-and-forward protocols and adaptive compression algorithms to handle intermittent connectivity exceeding 60 seconds duration."*

#### Original Requirement: 6.6 (Operational Accountability)

*"We require accountability for mission activities and operational decisions."*







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#### Interpreted Requirement: DATA-ACC-001

*"All mission activities, operational decisions, and system state changes shall be logged with millisecond timestamps, operator identification, and contextual metadata. Logs shall be retained for 7 years in WORM storage with cryptographic integrity verification, supporting regulatory compliance and post-mission investigation."*

#### 4.4 Categorize Activity

Interpreted requirements were organized into an SRS-compatible structure, establishing traceability and logical grouping.

| Category                            | Requirements Coverage                                                                             | Count |
|-------------------------------------|---------------------------------------------------------------------------------------------------|-------|
| <b>Safety &amp; Emergency</b>       | Autonomous emergency protocols, fail-safe states, abnormal condition handling, mission continuity | 12    |
| <b>Communication Systems</b>        | Bandwidth management, latency handling, intermittent connectivity protocols, secure communication | 8     |
| <b>Data Management</b>              | Operational logging, audit trails, data retention, post-mission review support                    | 7     |
| <b>Autonomous Operations</b>        | Mission execution, fleet coordination, remote oversight, autonomous decision-making               | 11    |
| <b>System Integration</b>           | Sensor interface integration, external system coordination, platform compatibility                | 9     |
| <b>Monitoring &amp; Diagnostics</b> | Vehicle status monitoring, operational integrity reporting, system health assessment              | 10    |

Table 5: Categorized Requirements for SRS Integration





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### Standardized Codification Scheme:

- **SYS-xxx:** System-level architectural requirements
- **SAF-xxx:** Safety-critical requirements
- **COM-xxx:** Communication subsystem requirements
- **DATA-xxx:** Data management and logging requirements
- **AUTO-xxx:** Autonomous operation requirements
- **INT-xxx:** System integration requirements
- **MON-xxx:** Monitoring and diagnostic requirements

All categorized requirements have been uploaded to our secure repository with full traceability to original PS clauses.

## 5 Incongruous Requirements Checklist Methodology

The Incongruous Requirements Checklist (IRC) methodology guided our systematic identification of PS incongruities. Our checklist implementation focused on:

### 5.1 IRC Validation Framework

- **Ambiguity Detection:** Systematic identification of subjective terminology requiring quantitative definition
- **Testability Validation:** Assessment of requirement verifiability through operational testing
- **Consistency Checking:** Cross-sectional analysis for requirement coherence and contradiction





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- **Completeness Verification:** Gap analysis against industry standards and operational context

## 5.2 Quantitative Analysis Results

- Total PS requirements analyzed: 57 clauses
- Incongruities identified: 15 (26.3% incongruity rate)
- Critical priority findings: 2 (13.3% of incongruities)
- High priority findings: 3 (20.0% of incongruities)

## 6 Risk Analysis and Mitigation

Our analysis identifies several implementation risks requiring proactive mitigation.

### 6.1 Critical Technical Risks

#### 6.1.1 Risk: Communication Specification Ambiguity

**Description:** PS communication constraints lack quantitative specifications, creating implementation uncertainty.

**Impact:** System may fail under actual deep-sea communication conditions (Severity: High).

**Likelihood:** Probable ( $\geq 60\%$  occurrence).

**Mitigation Strategy:** Establish quantitative communication requirements through prototype testing and stakeholder validation.

#### 6.1.2 Risk: Emergency Protocol Definition Gap

**Description:** Emergency handling requirements lack operational procedures and escalation protocols.





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**Impact:** Inconsistent emergency response could compromise mission safety (Severity: Critical).

**Likelihood:** Possible (40-60% occurrence).

**Mitigation Strategy:** Joint development of detailed emergency response protocols with Triton operational staff.

#### 6.1.3 Risk: Sensor Integration Complexity

**Description:** Sensor integration requirements lack interface specifications and protocol definitions.

**Impact:** Integration delays affecting project schedule and functionality (Severity: High).

**Likelihood:** Probable ( $\geq 60\%$  occurrence).

**Mitigation Strategy:** Early prototyping of sensor interfaces and development of Interface Control Documents (ICDs).

#### 6.1.4 Risk: Autonomy Level Misalignment

**Description:** Unmanned operations context lacks clear autonomy level definitions.

**Impact:** System may not properly balance autonomous operation with necessary human oversight (Severity: Medium).

**Likelihood:** Possible (40-60% occurrence).

**Mitigation Strategy:** Definition of autonomy levels and handoff protocols through operational scenario analysis.

## 6.2 Project Management Risks

### 6.2.1 Risk: Scope Creep Post-Clarification

**Description:** Stakeholder clarification may introduce requirements beyond original PS scope.

**Impact:** Budget and schedule overruns affecting project viability.





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**Mitigation Strategy:** Formal change control process managed through Account Manager escalation.

#### 6.2.2 Risk: Stakeholder Knowledge Attenuation

**Description:** Key stakeholder availability may decrease during implementation phase.

**Impact:** Loss of contextual knowledge affecting requirement interpretation.

**Mitigation Strategy:** Comprehensive documentation and knowledge transfer protocols.

## 7 Revised Requirements Summary

Our BCIC analysis has produced the following outcomes:

| Metric                                   | Value      |
|------------------------------------------|------------|
| Total PS requirements analyzed           | 57         |
| Incongruities identified                 | 15 (26.3%) |
| Requirements clarified and reinterpreted | 57 (100%)  |
| New atomic requirements created          | 22         |
| Requirements deprecated                  | 0          |
| Traceability coverage to PS clauses      | 100%       |

Table 6: Analysis Process Metrics

## 8 Repository Structure and Configuration Management

All requirements, analysis artifacts, and supporting documentation are maintained in our secure GitHub repository following industry-standard configuration management practices.





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### 8.1 Repository Architecture

- **Repository URL:** <https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113>
- **Access Protocol:** Role-based access control with audit logging
- **Primary Branch:** `main` - Contains all reviewed and approved content
- **Development Model:** Git Flow with protected branches

### 8.2 Configuration Management Features

- **Version Control:** Git-based change tracking with cryptographic commit signing
- **Branching Strategy:** Feature branches per requirement category with mandatory peer review
- **Issue Tracking:** Integrated issue management for incongruity tracking and resolution
- **Documentation:** Wiki with methodology documentation and process guidelines
- **Release Management:** Semantic versioning for analysis milestones

### 8.3 Traceability Implementation

The repository implements comprehensive traceability through:

- Bidirectional linking between PS requirements and interpreted requirements
- Automated validation scripts for requirement completeness and consistency
- Cross-reference matrices maintaining requirement lineage
- Digital signatures for requirement approval and change authorization





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## 9 Account Manager Engagement and Stakeholder Coordination

Brigman Deep Sea Technology's dedicated Account Manager facilitated the analysis process, ensuring effective stakeholder engagement and risk mitigation.

### 9.1 Account Manager Profile

- **Name:** Muhammad Ridwan Putra Jasni, Project Lead
- **Employee ID:** BDST-RE-001
- **Department:** Requirements Engineering
- **Role:** Primary liaison between vendor analysis team and customer stakeholders

### 9.2 Analysis Phase Contributions

| Activity Phase         | Specific Contributions                                                               |
|------------------------|--------------------------------------------------------------------------------------|
| <b>Breakdown</b>       | Facilitated stakeholder introductions; coordinated team analysis of PS incongruities |
| <b>Clarify</b>         | Scheduled and moderated stakeholder sessions; documented clarification outcomes      |
| <b>Risk Management</b> | Coordinated risk identification workshops with team specialists                      |
| <b>Communication</b>   | Managed communication protocols between vendor team and customer representatives     |

Table 7: Account Manager Contributions





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### 9.3 Escalation Protocol

A formal escalation protocol ensures timely resolution of technical and procedural issues:

1. **Level 1:** Technical team resolution (within 24 hours)
2. **Level 2:** Project Lead mediation (within 4 hours if unresolved)
3. **Level 3:** Vendor executive engagement (VP of Engineering)
4. **Level 4:** Contractual and scope review

## 10 Analysis Completion and Approval

This BCIC analysis has been completed by the Brigman Deep Sea Technology Requirements Analysis Team and is ready for SRS development commencement.

### TEAM APPROVAL AND SIGN-OFF







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| Team Member                 | Employee ID | Signature |
|-----------------------------|-------------|-----------|
| Muhammad Ridwan Putra Jasni | BDST-RE-001 |           |
| Chng Zong Han               | BDST-SE-015 |           |
| Lin Yuhao                   | BDST-CS-022 |           |
| Muhammad Mikhail Bin Mazlan | BDST-AI-029 |           |
| Kannan s/o Rajamohan        | BDST-DM-036 |           |

Table 8: Team Approval Signatures

**DATE:** February 7, 2026

**ANALYSIS VERSION:** 1.0

**STATUS:** Complete - Ready for SRS Development

#### PROJECT AUTHORIZATION

*This requirements analysis has been reviewed and approved by all team members.*

*The analysis is complete and ready for progression to the SRS development phase.*

